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Detecting Violent Arm Movements Using CNN-LSTM

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***Abstract*—With the increasing rate of surveillance camera utilization in household residences, intelligent security systems capable of automatically identifying violent activity has already become a significant research issue in recent times. Indoor violence mainly comes in the form of punching, beating and scuffling which primarily involve arm movements. In this paper, an end-to-end deep learning framework for detecting violent arm movements has been proposed. To consider the motion information of the violent movements from the surveillance video clips, a combination of Convolutional Neural Network (CNN) and Long-Short-Term-Memory (LSTM) model is used. The CNN model extracts the spatial features which are passed to the LSTM model for temporal feature learning. The spatio-temporal features are then fed to the Softmax classifier for binary classification and outputs violent or non-violent label. Experimental results obtained after evaluating the proposed model on the benchmark Hockey Fight dataset exhibit higher classification accuracy than some of the existing state-of-the-art methods, thus, proving superior performance in the context of detecting violent arm movements.**

***Index Terms*—violence detection, CNN-LSTM, indoor surveillance, arm movements, spatio-temporal features**

I. Introduction

Over the years, the use of surveillance cameras inside household residences has undergone significant increase due to the rise in the number of working parents. Due to the lack of parental supervision, child abuse inside residences has also increased in the past decade.

Manual human supervision is required in almost every existing indoor surveillance system to monitor the occurrence of any type of abnormal activity. This imposes that the existing manual systems for surveillance are not feasible due to the limitation of human resources. Since domestic violence occurs infrequently in most cases, an autonomous monitoring system, capable of automatically identifying and classifying violent and non-violent behavior would eliminate the need for manual supervision.

The main types of violent domestic activities include beating and scuffling which primarily involve the movement of arms and hands with very few leg movements involved in the main action. To identify these kinds of movements from residential surveillance include two main types of computer vision tasks – activity recognition or prediction and action classification. For identification of any action from static images, only the spatial information can do the job. However, predicting and action from video clips would also require the motion information along with the spatial information. Therefore, the accurate recognition of violent arm movements from videos require both spatial and temporal features. The existing handcrafted approaches used for detecting violent behavior from videos have complex structure and require elaborate pre-processing. The use of a deep-learning-based model, however, can significantly simplify the learning structure, effectively reducing the need for compound pre-processing by learning features from raw input data. Therefore, models based on deep learning can be used for similar violence detection tasks.

In this paper, an end-to-end model based on deep learning for binary classification to identify violent arm movements in residential context has been proposed. The model can effectively learn spatial and temporal information from feature maps gained from the raw input videos. The main contributions of our proposed method can be summarized as follows:

- An end-to-end trainable deep learning model has been proposed for performing the classification of violent arm movement
- Convolutional Neural Network combined with Long Short Term Memory is proposed to simplify pre-processing for raw video data
- The effectiveness of the proposed model has been validated using a widely used benchmark dataset for classification of violent video clips

The rest of the paper is organized as follows: Section 2 discusses the relevant works in the area of violent video detection. The details of the proposed model are explained in Section 3. It is followed by Section 4, which contains the details of implementation of the proposed architecture. The experimental results and the model's comparison with several state-of-the-art methods are discussed in Section 5 followed by Section 6, which concludes this paper.

II. Related Works

Researchers have proposed several methods to address the problem of detecting violence from video clips. Earlier methods incorporated the use of handcrafted features for violence detection. correspond to the attributes generated using the information present in the image itself using different algorithms. For example, the use of linear SVM can be seen in the research of Zhang et al. [1], who used Gaussian Model of Optical Flow (GMOF) to extract candidate violent regions

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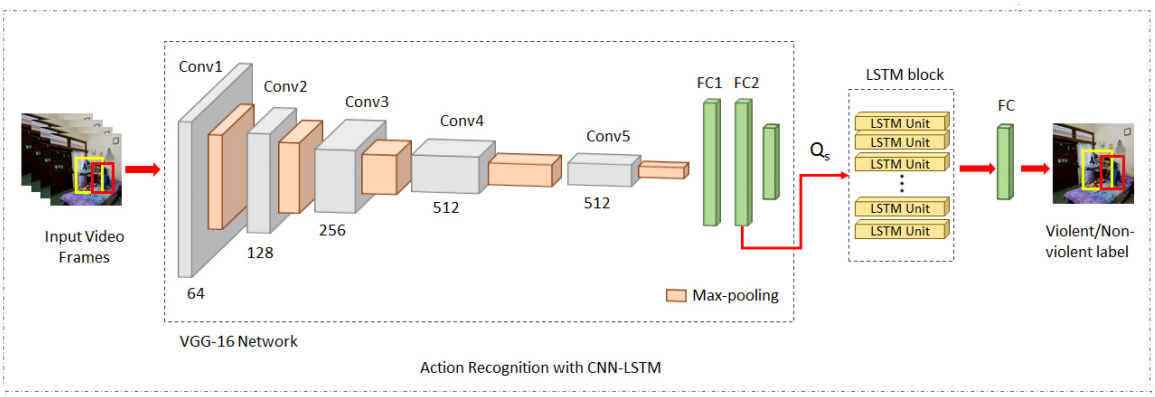


Fig. 1. Proposed network architecture for violent arm movement detection

followed by Orientation Histogram of Optical Flow (OHOF) descriptor to classify violent events. Oriented Violent Flow Descriptor (OViF) was propposed by Gao et al. [2] to resolve this problem, which is an enhanced version of ViF for non-crowded contexts. The Bag-of-Words (BoW) model coupled with Space-Time Interest Points (STIP) and Motion SIFT Interest Points (STIP) has been used by Bermejo et al. [3]. Peipei et al. [4] used Local Histogram of Oriented Gradient (LHOG) descriptor and the Local Histogram of Optical Flow (LHOF) descriptor and classified the extracted features using BoW model.

However, due to the computational complexity of hand-crafted features, deep learning approaches are increasing in practice in the area of recognition of human activity. Recent frameworks for violence detection have been focusing primarily on deep learning based approaches. of which, the work of Peixoto et al. focused on using a combination of 3D Convolutional Neural Networks (3D CNN) and CNN-LSTM for gaining knowledge of the spatial-temporal feature maps from the raw video inputs followed by a shallow Feature Fusion Network to predict the presence of violence in the sequence of frames [5]. A model based on VGG-19 and LSTM was used for real-time violence detection by Abdali et al. [6].

Song et al. improved the preprocessing method of the 3D CNN model, modifying it by incorporating a new sampling method using Grey Centroid algorithm as the key frame extraction algorithm instead of uniform sampling [7]. Sudhakaran et al. [8] proposed a model that uses 2D CNN and Convolutional Long-Short Term Memory (ConvLSTM) to capture spatio-temporal localized characteristics that enable the study of local motion taking place in the videos. His work was improved by Hanson et al. [9], who included bidirectional temporal encodings and element wise max pooling on the existing ConvLSTM model to build a Bidirectional Convolutional LSTM (BiConvLSTM) as spatiotemporal encoder. Sharma et al. [10], in their work, proposed to use a pre-trained ResNet-50 model for violence detection in videos. The use of Bidirectional LSTM can be seen in [11], [12] for violence detection. Multi-stream deep neural network approaches were also proposed as a method for violence detection [13]. Other approaches include the use of Temporal Segment Network(TSN) [14], Temporal Robust Features(TRoF) descriptor for movement detection [15], the combination of a Bidirectional Gated Recurrent Unit (BiGRU) and a 2D Convolutional Neural Network (CNN) [16]. In another approach, Khatun et al. used smartphone accelerometer sensor to collect data and CNN, SVM and Random Forest classifiers to detect different types of human movements [17].

Some of the most prominent recent techniques for violence detection in videos include the work of Asad et al. [18] who demonstrated the use of multi-frame feature fusion network to combine low and high level spatial features. However, use of two VGG-16 networks takes notably longer time to train. Ullah et al. proposed a method for violence detection based on C3D architecture that used MobileNet-SSD CNN model for person detection and 3D CNN for spatiotemporal feature extraction followed by a Softmax classifier to analyze the activity and produce a class label [19].

Although these techniques show significant results in violence detection in different contexts, however, no notable work has been done in the context of indoor residential surveillance particularly focusing on violent arm movements.

III. Proposed Methodology

To address the problem of identifying violent arm movements in the context of indoor residential surveillance, an end-to-end deep learning network is proposed in this paper. Actions in a video are described by the motion information in multiple frame sequences. Simple convolutional networks are inefficient in learning temporal information. But, by combining LSTM with CNN results in a network that can efficiently learn both spatial and temporal features. In this step, a combination of CNN and LSTM model is proposed for spatio-temporal feature extraction. The proposed network is mainly based on VGG-16, proposed by Simonyan et al. [20] which works as the spatial feature extractor followed by an LSTM block for temporal feature learning. Fig 1 illustrates the detailed network layout.

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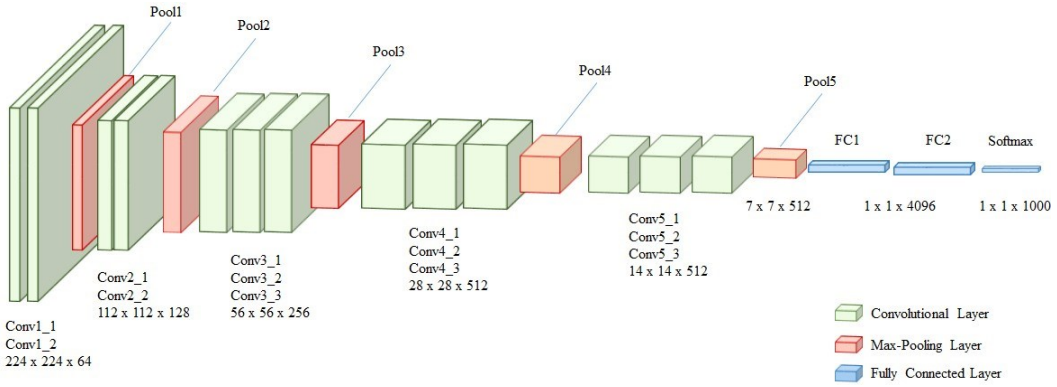


Fig. 2. VGG-16 network architecture

The proposed system starts by extracting frames from the input surveillance video clips then learning spatial information using VGG-16 followed by the process of learning temporal information using LSTM. Finally, a Softmax classifier is used to predict action classes as violent or non-violent. The following subsections discuss the details of internal steps of abuse detection. A set of 20 frames are extracted from the input video and sent to the following VGG-16 network for batch processing.

A. Extracting Spatial Feature using CNN

The VGG-16 network is a series of convolutional neural networks and fully connected networks that is used for large scale image processing. This network is basically used for prediction, feature extraction and fine tuning. The proposed system will use this model for transfer learning from the video frames and for classification. The transfer values are saved to a cache-file immediately preceding to the final classification layer of this model. While processing images more than once, caching the transfer-values save up a lot of time. Fig 2 illustrates the detailed architecture of the VGG-16 network. The network consists of 5 basic building blocks, followed by 2 fully connected networks at the end represented by *FC1* and *FC2* and one softmax prediction layer. Each block is a series of convolutional layers followed by a layer of max-pooling that executes spatial downsampling. The input layer to the last max pooling layer is the part that is responsible for feature extraction. The remaining layers are responsible for classification. This VGG-16 model will expect the shape of the input to be $224 \times 224 \times 3$.

B. Extracting Temporal Feature using LSTM

LSTMs are a special type of Recurrent Neural Networks(RNN). Recurrent Neural Networks are networks that have loops in them which allows information to persist. But, RNNs generally face the problem of handling long-term dependencies which is eliminated by the emergence of LSTMs. For long periods of time, LSTMs remember details, which allow them to acquire knowledge of long temporal features from the previous vectors of the VGG-16 function. In this way, from sequential input video frames, LSTM units allow the action recognition model to learn the internal representation of motion information over time.

The transfer values obtained from the VGG-16 network are fed to the LSTM block. After processing each video frame, the VGG-16 network produces an output vector having 4096 transfer values. Hence, processing 20 frames will give us 20×4096 transfer values per video.

C. Model Optimization

The proposed model has been optimized using Adam optimization algorithm. It is an addition to the traditional stochastic gradient descent procedure. This algorithm has been used for its computational efficiency with non-stationary objectives and small memory necessities also for its adeptness in handling large amount of video data. In Keras API with a Tensorflow backend, which is used for this model, the default value of the initial learning rate is set to 10^{-3}. The loss for the model is expressed as mean squared error which can be expressed as Eq. 1.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(F_i - \bar{Y}_i)^2 \quad (1)$$

Here, F_i is the actual output value and $\bar{Y}_i$ is the predicted output value of the model at point i. The total number of data points is n.

IV. IMPLEMENTATION

A. Pre-Processing

The action recognition model processes the video clips in batches of 20 frames. This model expects the resolution of the input frames to be 224×224. However, the dimensions of the the frames in the clips of the Hockey Fight dataset are 360×288. So, after downloading the dataset, first, frames are extracted from the video files and converted into a suitable

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TABLE I
STATISTICAL SUMMARY FOR HOCKEY FIGHT DATASET.

Dataset	Frame Size	Total No. of Videos	No. of Videos with Fight Scenes	Frame Rate (fps)	No. of Videos with Non-Fight Scenes	Frame Rate (fps)
Hockey Fights [3]	360 x 280	1000	500	25	500	25

format for the neural network. The dataset is separated into violent and non-violent categories. N number of frames, are obtained from each video clip. Then, the frames are resized into dimensions 224×224 for further processing. From each video, maximum 20 frames are picked. So here, *N=20*. If a video has more than 20 frames, the model selects frames at usual intervals by disregarding intermediate frames to keep away from repetitive computation. All the videos in the Hockey Fight dataset are around 2 seconds with 25fps giving a total of 50 frames per video. So, during pre-processing, only 20 frames are chosen for the training sample by leaving out each intermediate frame. This maximum number of frames was selected based on the video length in the dataset. A dataset with larger duration of clips would require a greater number of frames to be processed. No other frame processing has been done apart from converting the input videos into frames and resizing them according to the model requirements.

B. Experimental Settings

The architecture for the networks in the proposed approach has been implemented using OpenCV(Open Source Computer Vision) library and Keras API on Python in the Jupyter Notebook application. The project was run on Tensorflow backend, end-to-end. The experiments were carried out on NVIDIA GeForce MX230 GPU.

The VGG-16 model for spatial feature extraction was pre-trained with weights from the Imagenet dataset [21]. The instance of VGG-16 model is created using Keras API. This network is not trained at the time of the main training stage. Only the additional LSTM units have been trained on the input data from the Hockey Fight dataset. This model consists of of convolutional and max pooling layers. After that, there are 2 fully connected layers which is followed by a softmax output layer. The loaded model had a total of 138,357,544 parameters, all of which are trainable. This model processes the 20 frames from the video clip taken as input and extracts the spatial feature maps. The transfer values from the second fully connected layer, prior to the softmax layer, are saved in a cache file, to reduce the processing time required by VGG-16, if each image is processed more than once. Each frame produces a vector of 4096 transfer values and the VGG-16 network will process a maximum of 20 frames per video. So, each video produces a total of 81920 transfer values. These values are stored into disk and loaded into the RAM memory again prior to the training of the LSTM units.

The LSTM network used for the proposed approach has a total of 512 units. The input shape of the model is denoted as *(n_chunks, chunk_size)* where, *n_chunks* is the maximum number of frames to be processed, which is 20, in this case, and the *chunk_size* is the number of transfer values per frame, which is 4096. The initial learning rate is set to 10^{-3}. ReLU activation function is used in the sequential network. The network is initially trained for 20 epochs with a batch size of 100. The ADAM optimization function is used for backpropagation to minimize loss.

C. Hyperparameter Tuning

Hyperparameter tuning is done to achieve a stability between underfitting and overfitting. Underfitting occurs when the model is not capable to reduce the error for the test set or the train set or both. A model that is underfit, is not strong enough to fit the fundamental intricacies of the data distributions. The initial epochs and batch size were set to 20 and 100 respectively. These values are not optimal for the model. So, these hyperparameters have been tuned to 200 epochs and a batch size of 500, in order to achieve better performance.

D. Dataset

The Hockey Fight dataset, introduced by Nievas et al. [3] has been used to train and evaluate the proposed model. This dataset contains 1000 short video clips of ice hockey matches taken from the National Hockey League(NHL). In this dataset, 500 video clips contain fight scenes, and the rest of the 500 are non-fight videos. The frame rate for each video clip is 25fps. Each has length of around 2 seconds totalling into 50 frames per video clip with a resolution of 360×288 pixels. The statistics for the Hockey Fight dataset is summarized in Table I.

The videos labeled with fight class are all related to fights in the hockey grounds, and the non-fight class videos are also related to the same environment. So all the videos have similar background and human activities.

V. RESULTS AND DISCUSSION

The model is evaluated in terms of the performance of the LSTM network. Mean training, testing and validation accuracy and loss are used as the metrics to quantify the performance results of the experiments. The accuracy of the proposed technique for Hockey Fight dataset is evaluated using cross-validation method. The loss is calculated as mean squared error. The network parameters are selected on the basis of maximum test performance accuracy for the dataset. Initially,

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TABLE II
MAXIMUM ACCURACY AND LOSS STATISTICS FOR THE PROPOSED MODEL EVALUATED ON HOCKEY FIGHT DATASET

Set	Before Hyperparameter Tuning		After Hyperparameter Tuning	
	Accuracy(%)	Loss	Accuracy(%)	Loss
Training	92.5	0.0625	**96.4**	0.0361
Testing	92	0.0729	**92.5**	0.0570
Validation	96	0.0379	**98**	0.0236

the maximum epochs that the dataset is run for and batch size were 20 and 100 respectively. After hyperparameter tuning, the maximum number of epochs is set to 200 and the batch size is set to 500. Table II summarizes the training, testing and validation accuracy and losses before and after hyperparameter tuning.

Table II depicts that the proposed model achieved 96.4% training accuracy, 92.5% testing accuracy and 98% validation accuracy after tuning the hyperparameters. Fig.3(a) and Fig.3(b) represents the training and validation accuracy and losses for the overall lstm model on the Hockey Fight Dataset. The graph shows that the maximum training accuracy of 96.4% was achieved on the 192nd epoch. The maximum validation accuracy of 98% was achieved. The corresponding model loss for the highest testing accuracy, which was 92.5%, was 0.0570.

A. Discussions

From Table II, it can be seen that, before hyperparameter tuning, the training, testing and validation accuracy was 92.5%, 92% and 96% with loss values of 0.0625, 0.0729 and 0.0379 respectively. After the tuning process, when the maximum number of epochs were increased 10 times and the batch size was increased 5 times, the lstm model showed increased accuracy for all the sets and decreased losses. The training, testing and validation accuracy after hyperparameter tuning is 96.4%, 92.5% and 98% with loss values of 0.0361, 0.0570 and 0.0236 respectively. So, it can be said that tuning the epoch and batch size increased the overall performance of the abuse detection model.

The decreased loss after hyperparameter tuning implies that the model has been optimized well for the specific purpose. The classification accuracy increased from 92% to 92.5% after the tuning process which implies that the model fit the data well, thus, avoiding the chances of underfitting. It also implicates that the amount of data was sufficient to build an accurate model. However, the increase in performance accuracy was only 0.5%, which can be further improved by tuning the number of epochs and batch size with other hyperparameters too, such as the number of recurrent units, learning rate and by adding dropout probability.

B. Comparative Analysis

The performance of the proposed model compared against different state-of-the-art handcrafted and deep learning approaches is depicted in this section. The comparative models used in this analysis have all been evaluated on various datasets including the Hockey Fight dataset [3]. However, this analysis only compares the classification accuracy received for the Hockey Fight dataset. Table III compares the classification accuracy of the proposed model with these models on Hockey Fight dataset.

TABLE III
COMPARISON OF CLASSIFICATION ACCURACY OF THE PROPOSED MODEL WITH SOME EXISTING MODELS

Method	Classificaton Accuracy
OViF [2]	87.5%
MoSIFT+BoW [3]	88.8%
MoSIFT+HIK [3]	90.9%
HOG + BoW [3]	88.77%
Gracia et al. [22]	82.4%
Deniz et al. [23]	90.1%
RestNet-50 + ConvLSTM [10]	87.5%
AMDN [24]	89.7%
Proposed	**92.5%**

The handcrafted-based approaches that the model is compared against include: Oriented Violent Flow (OViF) [2], which is based on ViF descriptor with motion magnitudes and orientations between flow vectors, combination of different feature descriptors for action recognition such as Motion Scale invariant Feature Transform(MoSIFT) and Histogram Intersection Kernel(HIK), combination of MoSIFT with Histogram of Oriented Flow (HOF), combination of Histogram of Oriented Gradients(HOG) with Bag of Words(BoW) [3]. All of these combinations used BoW approach for violence detection.

The use of motion blob detection for feature extraction [22] and the integration of acceleration patterns through computing power transformations of two consecutive video frames to detect motion changes that are unanticipated [23] are other handcrafted comparative models. For comparative analysis, the methods which are based on deep learning include the use of ResNet-50 and ConvLSTM for violence detection in [10] and the use of Stacked Denoising Autoencoder(AMDN) in [24].

The above mentioned models are all widely used state- of-the-art models for violence detection in videos. In terms of classification precision, the Table III indicates that the proposed model surpasses these state-of-the-art models.

VI. CONCLUSIONS AND FUTURE WORK

This paper has proposed an end-to-end deep learning architecture for detecting abusive activity in the form of arm movements in surveillance videos. The model detects the violent arm movements in surveillance videos by learning spatial feature maps of the input frames using VGG-16 and

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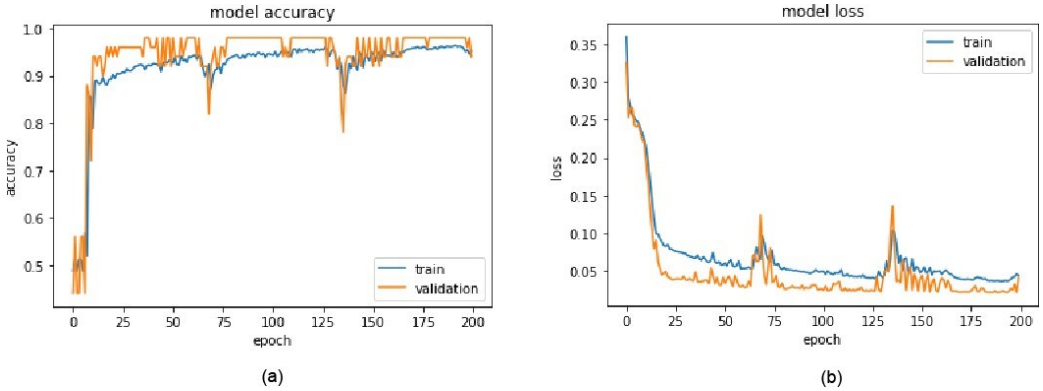


Fig. 3. (a) Overall LSTM model training and validation accuracy, (b) Training and validation loss

learning the temporal feature maps using Long Short-Term Memory units. The Softmax classifier does the final prediction with these extracted spatiotemporal features. Using a pre-trained model for transfer learning has enhanced the performance of the model and significantly reduced the training time. Temporal features were learned by the LSTM blocks to classify images based on the transfer-values from the VGG16 model. The proposed approach is evaluated on a standard benchmark dataset. Experimental results have demonstrated that this model is effective for detecting abusive activity from surveillance video datasets and outperforms many of the existing state-of-the-art methods for violence detection in videos and that deep learning methods provide more accurate detection of abusive activity than handcrafted methods. Future work will be dedicated to developing the combination for use in crowd contexts and also in improving the classification accuracy.

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